

Home SOC Lab & A.E.G.I.S. Cyber Ops Platform

Production-Grade SIEM, Multi-Cloud Detection Engineering & AI Command Center

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Live Deployment: <https://yashh127.github.io/home-soc-lab-aegis-cyber-ops/> | GitHub Repo: <https://github.com/yashh127/home-soc-lab-aegis-cyber-ops>

EXECUTIVE OVERVIEW

The **Home SOC Lab & A.E.G.I.S. Cyber Ops Platform** is an end-to-end, enterprise-grade Security Operations Center (SOC) infrastructure built to simulate real-world SIEM telemetry monitoring, detection engineering, and automated incident response. The environment ingests live and replayed security events from Active Directory Domain Controllers, Linux systems, and AWS Cloud (CloudTrail & GuardDuty), applying 20 custom correlation rules aligned directly with the MITRE ATT&CK framework.

SYSTEM ARCHITECTURE & CORE CAPABILITIES

1. Containerized SIEM Infrastructure

Deploys a containerized Wazuh 4.9 SIEM, OpenSearch Indexer, and Filebeat pipeline via Docker Compose. Normalizes multi-source telemetry over Syslog (UDP 514).
2. Detection Engineering & MITRE Rules

Features 20 custom XML correlation rules for Active Directory attacks (Kerberoasting T1558), Windows Sysmon MimiKatz dumping (T1003), Linux SUID escalation, and AWS root logins without MFA.
3. A.E.G.I.S. CYBER OPS Command UI

A React 18 + Vite wallboard featuring a J.A.R.V.I.S. AI Voice Dispatcher (Web Speech API), Global Threat Trajectory Map, Security Advisory Radar, and 1-click active response playbooks.

TECHNICAL SPECIFICATIONS & DETECTION CATALOG

Component	Technologies Used	Key Detections & Capabilities
SIEM & Database	Wazuh 4.9, OpenSearch 2.13, Filebeat, Docker Compose	Real-time log correlation, OpenSearch indexing, TLS encryption
Log Telemetry	Active Directory, Sysmon, Auditd, AWS CloudTrail, GuardDuty	Kerberoasting, MimiKatz lsass.exe, Root MFA abuse, S3 leaks
Command Center UI	React 18, Vite 8, Tailwind CSS, Web Speech/Audio APIs	J.A.R.V.I.S. voice alerts, Global Threat Map, Interactive JSON simulator
Automation & IR	Python 3 (log_replay.py), Bash (attack_simulation.sh)	1-click IP null-routing (iptables), AWS IAM key revocation, /tmp purge

PROJECT HIGHLIGHTS FOR EVALUATION

- Production-Ready Deployment:** Fully reproducible multi-container security stack initialized via single-command Docker Compose orchestration.
- MITRE ATT&CK Mapping:** 20 custom correlation rules created across 4 threat vectors with decoded JSON payload validation.
- Voice-Enabled IR Playbooks:** Interactive React command center with J.A.R.V.I.S. speech alerts and automated active response scripts.
- Open-Source Repository:** Source code, rules, decoders, and documentation published under MIT License on GitHub.